

A/B/G/P Pre-Freeze Design & Implementation Review v2

Metaloic Labs - provisional design review for text-implementation consistency before any scientific freeze.

Status: REVIEW_PENDING **Harness:** statistical/synthetic-fixture checks passed

End-to-end implementation qualification: PENDING **Confirmatory namespace:** untouched

Scope correction

The earlier v1 review used QUALIFIED too broadly. The verified 200-test / 21-fixture result established that the statistical analyzers, frozen synthetic challenge fixtures, exact-reference checks, replay, and namespace firewall behaved as specified. It did not establish that every experimental arm was implemented end to end. This v2 package preserves the agreed science and makes the remaining implementation and methodological blockers explicit.

Agreed scientific criteria - unchanged

Arm	Agreed target	Key freeze condition
A	Non-identifying verifier information beyond matched ordinary updating	Exact constructor-visible information budget; Bayesian-optimal control sees same permitted observations/messages; ≥ 5 pp plus arm significance and hard gates.
B	Capability/action-order recovery across genuinely different grammars	12 ordered directions; independent grammar machinery; held-out structural interventions; $> 90\%$ protected-order agreement and ≥ 15 pp advantage over required controls.
G	Dose response specific to action-relevant distinctions	Preclassified relevance; matched corruption; world-level blocked randomization; ≥ 15 pp max-dose gap plus corrected significance.
P	Persistence without verifier or reconstruction	Retained object only across hard restart; label-free invocation; zero verifier/search/reconstruction; ≥ 5 pp over strongest baseline; deletion within 2 pp of cold and preregistered reacquisition.

What the existing harness evidence supports

Existing DEV/QUAL evidence supports analysis mechanics: exact paired tests and Holm handling, B IUT plumbing, world-blocked G DP/reference agreement, frozen positive/ordinary-explanation/INVALID fixtures, deterministic replay, and confirmatory-namespaces isolation. These are harness checks only; they are not A/B/G/P scientific results.

Requirement-to-evidence status

Requirement	Current evidence	Status before freeze
A/P earliest shared stochastic ancestor	Episode/future seed separation and P restart mechanics exist. Complete ancestry audit over upstream sampled worlds, grammars, pools, and constructor state not yet established.	OPEN - audit required
A exact information-matched Bayesian baseline	Analysis/fixture path exists; end-to-end oracle and full PASS-event implementation still require verification.	OPEN - implementation/evidence
B 36-component IUT	IUT analysis path implemented. Current DEV B generator still assigns recovered order from target order; posterior/bisimulation control remains mechanics-only.	OPEN - real recovery/control paths
B alpha provenance	$\alpha = .0125$ is the conservative four-arm Holm component floor, not a multiplicity correction across B's 36 IUT components.	CLARIFIED
G world-blocked statistic	One world-level sign exchange across doses and exact DP/brute-force numerical agreement are implemented.	MECHANICS CHECKED
G exchangeability	Need design-level argument that full within-world vector is invariant under joint label exchange under H_0 , including generation, selection, stopping.	OPEN - justification required
P restart/deletion boundary	Independent acquisition episodes and restart/deletion mechanics exist in DEV/QUAL; earliest-ancestor independence and end-to-end ordinary-control paths still need audit.	OPEN - audit/implementation
Power for complete PASS event	Current B power includes 36-component IUT aggregation. A/P figures are paired-test calculations and do not establish complete multi-control plus hard-gate PASS power.	OPEN - recompute after final implementation

Bill Bao review points - 17 September

1. A/P independence must be established at the earliest shared stochastic ancestor, not inferred from distinct future seeds or hard restarts.
2. B IUT is defensible if PASS genuinely requires all 36 component alternatives. The role of $\alpha=.0125$ must be stated as coming from the four-arm familywise procedure, and power must be recomputed for the complete PASS event once real recovery and posterior/bisimulation controls are implemented.
3. For G, DP/brute-force agreement validates the calculation, not the exchangeability assumption. Freeze requires a design-level argument that the full within-world outcome vector is invariant to joint label exchange under H_0 , including generation, selection, and stopping.

Current power numbers - corrected interpretation

Arm	Candidate N	Reported calculation	Correct interpretation
A	4,096 episodes	0.825254	Paired-test power at 5 pp over declared discordance envelope; NOT complete-arm PASS power.
B	1,015/direction = 12,180	0.800598	Dependence-agnostic lower bound for simultaneous rejection across 36 IUT components; revisit with real controls and all gates.
G	421 worlds	0.801242	Conservative max-dose-only power under world-blocked paired model; exchangeability justification remains pending.
P	4,096 episodes	0.825254	Paired-test power at 5 pp over declared discordance envelope; NOT complete-arm PASS power.

Freeze rule

Do not mark the study FROZEN until every agreed requirement maps to a concrete implementation path and auditable evidence, the remaining methodological assumptions are justified, and power is recomputed for the actual complete PASS rules. Any change to hypotheses, effect floors, source-distinctness, inferential units, or PASS criteria must return to joint scientific review rather than being relabeled as an implementation clarification.

No confirmatory result is represented here. The confirmatory namespace remains outside this review package.